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Variable response of coastal environments of the northwestern Gulf of Mexico to sea-level rise and climate change: Implications for future change

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ABSTRACT

The results from nearly three decades of marine geological research in the northwestern Gulf of Mexico are compiled in an effort to understand those factors (e.g., sea-level rise, sediment supply, subsidence, antecedent topography) that influenced coastal evolution during the last eustatic cycle (~120 ka to Present). Armed with this information, we evaluate coastal response to variable sea-level rise of the Holocene and accelerated rise during historical time to gain a better understanding of how the coast is likely to respond to future changes.

The early Holocene evolution of northwestern Gulf of Mexico bays was punctuated by rapid and possibly synchronous flooding events that are interpreted as resulting from episodes of rapid sea-level rise. Two of these events, one between ~8.4 and 8.0 ka and the other between ~7.4 and 6.8 ka, were associated with known episodes of ice sheet retreat in North America and Antarctica, respectively.

During the middle and late Holocene, the east Texas and western Louisiana coasts experienced episodes of stability and growth followed by rapid shoreline retreat, while the central Texas coast remained relatively stable. This variability in coastal response to sea-level rise resulted mainly from differences in sediment supply and the highly irregular antecedent topography on which coastal environments formed. Sand that nourished the evolving east and south Texas, as well as westernmost Louisiana, coasts was derived mainly from transgressive ravinement of deltas that were formed during the falling stage of sea level (MIS5e–MIS2). The loci of these deltas controlled the spatial variability of this offshore sand supply. Sand supply to the central Texas coast has been dominated by converging longshore currents and throughout the middle to late Holocene was large enough to keep pace with sea-level rise. Moreover, sand supply from rivers has varied considerably in response to climate change. During the early Holocene, when the average rate of linear sea-level rise was 4.2 mm/yr, the Brazos, Colorado and Rio Grande Rivers all formed deltas on the inner continental shelf. Today, only the Brazos contributes enough sediment to the Gulf of Mexico to form a delta, but its sand delivery is minimal.

A reversal from late Holocene growth of western Louisiana–east Texas and south Texas barrier islands and peninsulas to erosion occurred during the last ~2.0 ka, after the rate of sea-level rise began to decelerate. The actual timing of this reversal was diachronous across the coast and was caused by a decrease in sediment supply once offshore sand sources were bypassed by the landward-advancing shoreface (transgressive ravinement). Differences in the exact timing and rate of retreat also reflect variability in the offshore profiles of these barriers and their thickness, which results from the underlying relief on the Pleistocene surface. The late Holocene was also the time of chenier plain development in western Louisiana and east Texas that was marked by localized differences in growth and retreat.

The current rate of eustatic sea-level rise in the northern Gulf of Mexico is in the range of 2.0 and 3.0 mm/yr and is approaching rates of the early Holocene. This is on average about five times the long-term rate of the past 4.0 ka. This larger rate is associated with an increase in coastal erosion in historical time relative to long-term rates. Current rates of Gulf shoreline erosion are approaching rates of the early to mid-Holocene and the rate of retreat of some barriers is not sustainable. Core transects off central Texas reflect a late Holocene history of shoreface progradation, but the modern shoreline is currently eroding, albeit at slower rates than elsewhere along the coast.

Overall, the sand supply needed to sustain the Gulf shoreline is lacking. Likewise, baylines are experiencing erosion rates that are faster than the late Holocene average and bayhead deltas are at the tipping point

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of catastrophic retreat. Spatial variability in erosion rates highlights the importance of factors such as sediment supply, subsidence, and anthropogenic influences on coastal evolution. Tropical storms and hurricanes, while exerting the most noticeable coastal change, merely exacerbate overall shoreline retreat and migration.

Along the entire northwestern Gulf Coast, the impact of human intervention to natural processes is particularly evident, primarily due to alterations in river discharge and sediment supply to the coast, interruptions in alongshore and cross-shore sand transport, and increases in rates of subsidence through fluid withdrawal. Compared to natural changes that took place during the Holocene, the impacts of human activity have been virtually instantaneous.

The geological record indicates that the ongoing increase in the rate of sea-level rise, coupled with diminished sediment supply and human intervention, will continue to severely impact the low-gradient coastal environments of the northwestern Gulf of Mexico.

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1. Introduction

The most recent scientific results from satellite altimetry indicate that the rate of global sea-level rise averages ~3.0 mm/yr, although the actual rate varies regionally (Church et al., 2004; Rahmstorf et al., 2007; Carlson, 2011; Church and White, 2011; NRC, 2012). However, within the northwestern Gulf of Mexico, subsidence exacerbates relative sea-level rise with rates as high as ~10 mm/yr, although the current rate of rise is hard to constrain because of considerable variability in subsidence across the region (Paine, 1993; Kolker et al., 2011; Simms et al., 2013). Regardless of the actual value, this is a significant increase over the long-term rate of the past few thousand years of ~0.40 to 0.60 mm/yr (Törnqvist et al., 2004a, 2004b, 2006; Simms et al., 2007b; Milliken et al., 2008a; Livsey and Simms, 2013). The evidence for accelerated sea-level rise in the northwestern Gulf of Mexico is consistent with results from the east coast of the US, which show that the rate of rise since the 19th century has significantly increased relative to the past few millennia (Engelhart et al., 2011; Kemp et al., 2011). It is generally accepted that the rate will continue to accelerate during the 21st century, but the actual magnitude of rise remains problematic due to uncertainty about the contribution of water from the Greenland and Antarctic Ice Sheets. An increased contribution appears more and more likely as our understanding of the state of these ice sheets increases (e.g. Rignot et al., 2011).

Current predictions about the impact of rising sea level on coasts rely largely on simple inundation models where the landscape elevation is flooded as sea level rises (e.g. NOAA, EPA). These models fail to capture the complex response of coastal systems to sea-level rise because they ignore changes in sediment accommodation, accretion, and erosion. There is general agreement among coastal scientists that we are still ill prepared to predict coastal response to accelerated sea-level rise and to other factors related to Global Climate Change, such as variations in sediment supply and changes in the frequency and magnitude of severe storms (Anderson et al., 2013a).

Here, we use results from marine geological research conducted during the past three decades that investigated the nature of Holocene sea-level rise and climate change and their impacts on the Northwestern Gulf Coast (NGC) (Fig. 1). We describe the spatial and temporal variability in coastal response to different rates of sea-level rise and provide context to changes that are occurring along the coast today. We further address the evidence for punctuated sea-level rise and its impacts. While this is a regional study, our message is that the response of coastal systems to Global Climate Change is controlled by multiple factors, and thus exhibits considerable spatial and temporal variability. We use the operational definition of the Holocene as the period of geologic time after the Younger Dryas cold event (~11.5 ka to present).

2. Data set

Over the last three decades, we have acquired over 10,000 km of high-resolution seismic data from the continental shelf using the Rice University research vessel *R/V Lone Star* and within area bays using the *R/V Matagorda*, *R/V Trinity* and the University of North Carolina's *R/V Jenny* (Fig. 1). Seismic data from the bays are augmented by data collected by the United States Geological Survey, mostly during the 1970s. The majority of the high-resolution seismic profiles from the continental shelf were digitally collected using a 15 in^3 water gun, with limited post-cruise processing (see Anderson et al., 2004 for details). Chirp and boomer profiles were taken within bays. Several hundred pneumatic-hammer cores and vibracores were collected and analyzed. More than 70 rotary drill cores (up to 25 m long) were acquired within the bays using the *R/V Trinity* (see Anderson and Rodriguez, 2008; Simms et al., 2010; Troiani et al., 2011).

A robust chronostratigraphic framework consisting of several hundred radiocarbon dates, mostly AMS, were used to constrain the ages of sediment and timing of events. More details about the methods used, including calibration of radiocarbon ages, can be found in Simms et al. (2007b, 2009a) and Milliken et al. (2008a, 2008b, 2008c).

3. Physiographic, oceanographic and climatic setting

For about 600 km along the northwestern Gulf of Mexico, the modern shoreline includes extensive coastal wetlands and bays, chenier plains, barrier islands and peninsulas, the Brazos wave-dominated delta and several tidal inlet/delta complexes (Fig. 1).

The shoreline is sculpted by fair-weather waves in the near shore that typically range from 30 to 60 cm high, with periods of 2 to 6 s and a diurnal, microtidal range (<1 m) (Morton, 1994). Prevailing winds are out of the southeast with more variable winds during the passage of low-pressure systems from the west and north. Because of the curved shape of the coastline, prevailing southeasterly winds cause longshore currents to flow from east to west in east Texas and from south to north in south Texas. This results in a longshore current convergence zone offshore of central Texas (Lohse, 1955; Curray, 1960; Morton, 1979). The area is subject to frequent severe storms. Over the late Holocene, intense hurricanes (likely category 3 and higher) have impacted the Texas coast at a time-averaged rate of 0.46% (annual landfall probability) (Wallace and Anderson, 2010).

From onshore to offshore, the low-gradient coastal plain transitions to a continental shelf that exhibits variable physiography in terms of width and gradient (Fig. 1). The shelf physiography is largely controlled by order-of-magnitude differences in long-term river sediment supply, which can be linked to differences in river drainage-basin area and climate variability (Anderson et al., 2004).

The area currently includes four major climate regimes (Thornthwaite, 1948): humid (western Louisiana and far east Texas), wet subhumid (east central Texas), dry subhumid (central Texas), and semiarid (south Texas). Holocene climate records indicate that these climate zones have shifted in the past between warm-arid and cool-moist at millennial time scales, although the exact cause(s) of these climate changes is not fully known (Toomey et al., 1993).

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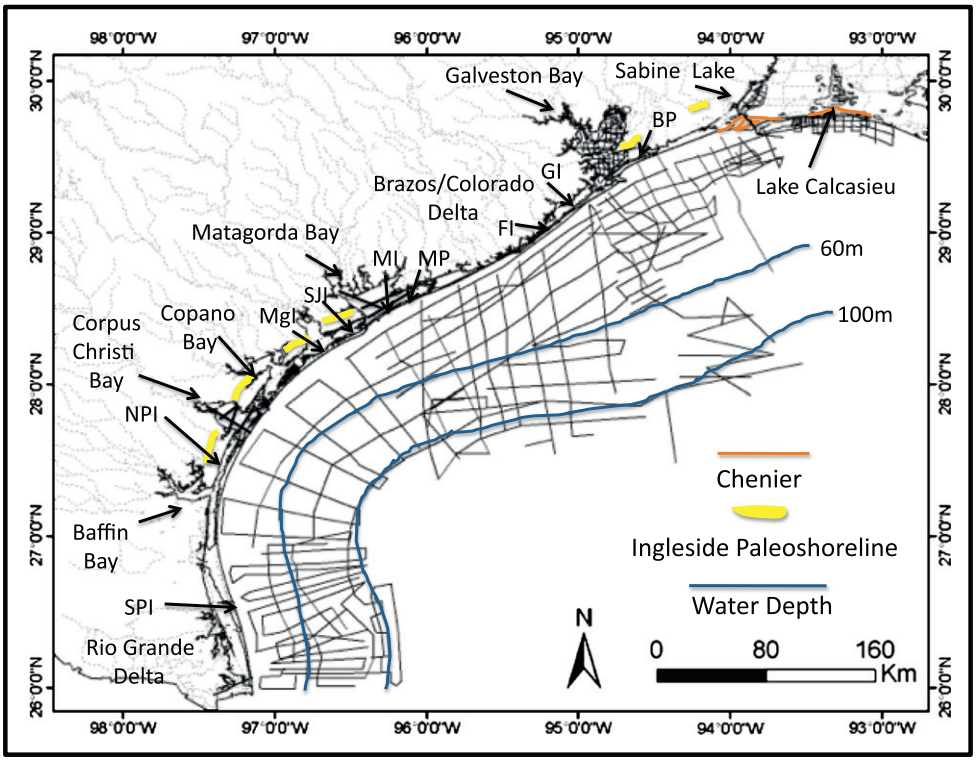

Fig. 1. Map of the northwestern Gulf of Mexico showing locations of places and features referred to in text. Also shown are track lines for offshore and nearshore seismic profiles used for this study. Ingleside paleoshoreline locations from Simms et al. (2013). BP = Bolivar Peninsula, GI = Galveston Island, FI = Follets Island, MP = Matagorda Peninsula, MI = Matagorda Island, SJI = San José Island, MgI = Mustang Island, NPI = North Padre Island, and SPI = South Padre Island.

Dams exist on all the major rivers and several rivers have been either diverted (e.g. the Brazos and Colorado Rivers) or their discharge so heavily diminished that their sediment supply has been severely depleted (e.g. the Rio Grande River). Dredging and construction of jetties at the mouths of bays has severely altered their natural tidal circulation and sedimentation. In addition, numerous beach and dune restoration projects along the coast have been marginally successful and some have adversely impacted the natural coastal ecosystem by mining sand from the active shoreface and the central barriers, or constructing unsustainable dunes with geotubes that rapidly become exposed and are inhospitable for vegetation (Wallace et al., 2009).

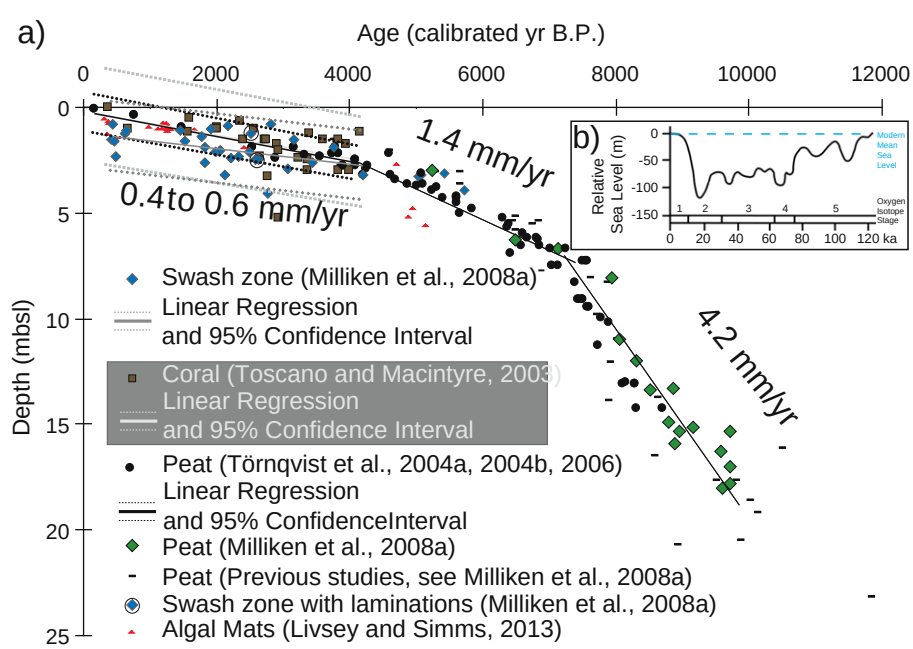

Fig. 2. Sea-level curve (a) for the northwestern Gulf of Mexico from 10 ka to Present (modified from Milliken et al., 2008a) and (b) for 120 ka to Present (modified from Chappell and Shackleton, 1986; Chappell et al., 1996; Autin and Aslan, 2001). For error bars, see referenced studies. Note, Livsey and Simms (2013) is not included in the linear regression analyses in (a). Data from Toscano and Macintyre (2003).

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4. Sea level and subsidence setting

Relative sea-level rise (RSL) in the northwestern Gulf of Mexico has been well constrained for the middle and late Holocene (Törnqvist et al., 2004a, 2004b, 2006; Simms et al., 2007b; Milliken et al., 2008a; Livsey and Simms, 2013). These studies show a progressive deceleration in the average linear rate of rise from 4.2 mm/yr to between 0.6 and 0.4 mm/yr over the past 10.0 ka (Fig. 2). The historic average rate of relative sea-level rise along the Texas coast decreases from 6.4 mm/yr (Galveston Island) to 5.2 mm/yr (San José Island) to 3.5 mm/yr (South Padre Island) (NOAA, 2012). These differences are a result of the sum of eustatic sea-level rise, local variations due to glacial-isostatic-adjustment-induced changes, local steric changes, local oceanographic influences, and local and regional subsidence (Morton et al., 2006; Simms et al., 2007b; Kolker et al., 2011; Paine et al., 2012; Simms et al., 2013).

Paine (1993) made one of the first estimates of subsidence rates for the Texas coast over geologic timescales that indicated rates <0.1 mm/yr. More recently, Simms et al. (2013) calculated long-term subsidence rates for the Texas coast based on the differential elevations of modern barrier islands and their marine isotope stage 5e (MIS5e) counterparts. Their results generally support the low rates of Paine (1993), but show spatial variability with increasing subsidence rates seaward and at locations closer to the Brazos/Colorado deltaic headland. The lowest rate of subsidence was ~0.03 mm/yr at the most inland site and the highest rate was ~0.09 mm/yr near the modern Brazos Delta. This spatial pattern of subsidence suggests that most of the long-term vertical motion is due to sediment loading. These long-term rates are two orders of magnitude less than modern rates of subsidence, derived by comparing tide gauges along Texas to the gauge in Pensacola, FL where subsidence is minimal due to shallow carbonate-rock substrate (Kolker et al., 2011). Historical subsidence rates in Texas are caused by shallow natural and anthropogenic processes (e.g., consolidation and fluid withdrawal), which vary locally and range between ~3 and 22 mm/yr (Paine, 1993; Morton et al., 2006; Kolker et al., 2011).

5. Pre-Holocene coastal evolution (MIS5e–MIS2)

While this paper focuses on coastal evolution during the Holocene, it is important to look further back in time to gain a complete understanding of those events that laid the foundation for later coastal evolution by shaping the relief on which Holocene coastal environments formed and by delivering sand that would ultimately nourish the modern cheniers, coastal barriers and peninsulas.

Marine Isotope Stage 5e (MIS5e) of ~120 ka represents the last interglacial and is viewed by some models as a likely sea level and associated coastal configuration for a Greenhouse World (e.g. Otto-Bliesner et al., 2006; Dutton and Lambeck, 2012). The volume of water in the oceans during this time period was equivalent to sea levels ~5–9 m above present (Cuffey and Marshall, 2000; Lambeck and Chappell, 2001; Kopp et al., 2009; Dutton and Lambeck, 2012; Lambeck et al., 2012; Kopp et al., 2013).

In Texas and western Louisiana, there is a prominent elongate ridge, known as the Ingleside Paleoshoreline, of approximately 3–5 m elevation that extends from eastern Louisiana to just south of the city of Corpus Christi, Texas (Fig. 1). This feature has long been considered to represent the last interglacial highstand (Price, 1933; Shepard and Moore, 1955; Morton and Price, 1987; Paine, 1993; Otvos and Howat, 1996; Otvos, 2005), although direct age control on this feature initially provided mixed results. Simms et al. (2013) obtained six optically stimulated luminescence ages of features long considered to be MIS5e. The ages cluster into three groups of 119 ka–127 ka BP, 54 ka–57 ka, and 1.3 ka. The two younger groups of ages correspond to regional periods of eolian reworking of sediment (Otvos, 2004; Forman et al., 2009). Thus, the ages of 119–127 ka for the core of the geomorphic feature support earlier interpretations that the Ingleside Shoreline is a MIS5e shoreline deposit.

For the most part, offshore paleoshoreline locations for the MIS5 through MIS2 period of sea-level fall are poorly constrained. One exception is the MIS5a paleoshoreline, which occurs at -11 ± 2 m water depth on the east Texas continental shelf (Simms et al., 2009b). These same authors refine earlier estimates (Rodriguez et al., 2000b) and suggest that the MIS3 paleoshoreline may have been situated as high as -20 m after correcting for local glacio-hydro-isostatic effects.

During the overall fall in sea level between MIS5 and MIS2 (Fig. 2), rivers with larger drainage basins, including the Red, Brazos, Colorado, Rio Grande and Mississippi Rivers (Shen et al., 2012), formed extensive deltas on the continental shelf (some are shown in Fig. 3). These deltas filled available accommodation, resulting in broad, low-gradient continental shelves offshore south Texas and east Texas and western Louisiana (Berryhill and Suter, 1986; Anderson et al., 1996; Abdulah et al., 2004; Banfield and Anderson, 2004; Wellner et al., 2004; Fig. 1). The formation of these fluvial-dominated deltas implies larger sediment supply of rivers relative to today (Anderson et al., 2004). This is attributed to the relatively rapid rate of sea-level fall, slow subsidence, and channel-belt formation on the inner shelf that resulted in erosion and recycling of sediments to the outer shelf. In addition, purging of sediment from fluvial valleys during sea-level fall contributed tens of cubic kilometers of sediment to the coast. In contrast, rivers with smaller drainage basins, such as those in central Texas, were unable to fill accommodation created by subsidence, resulting in a ramp-like continental shelf profile (Eckles et al., 2004). However, sand supply to the central Texas coast by converging longshore currents nourished an extensive strandplain that prograded across the inner shelf during the falling stage of sea level. As the overall rate of sea-level fall increased toward the end of the falling stage (post-MIS3; Fig. 2), sand supply could not keep pace with the rate of fall and shoreline progradation terminated at the mid-shelf.

Throughout the falling stage of sea level and subaerial exposure of the continental shelf, rivers formed extensive valleys defined by an exposure surface with considerable relief (Simms et al., 2007a; Fig. 4). This exposure surface is the framework on which modern coastal systems would be constructed. Furthermore, the eustatic lowstand was a time when significant sand bypass took place, especially in the case of smaller rivers like the Trinity River, where most of the sand being transported by the river during the falling stage was confined to the valley as opposed to being delivered to deltas. Hence, purging of these valleys resulted in much of the sand transported by these rivers bypassing the shelf (Anderson et al., 2004).

Detailed studies of the Calcasieu, Sabine, Trinity, Brazos, Colorado, Lavaca, Copano, Nueces, Navidad, and Baffin Bay system incised river valleys (Simms et al., 2006b; Anderson et al., 2008; Maddox et al., 2008; Milliken et al., 2008b, 2008c; Taha and Anderson, 2008; Simms et al., 2010; Troiani et al., 2011) included lithofacies mapping of valley-fill deposits and radiocarbon dating to constrain bay evolution. This work showed that these valleys were occupied for more than one eustatic cycle (compound valleys; Zaitlin et al., 1994), which explains why on the continental shelf even smaller rivers, like the Lavaca and Nueces Rivers, have valleys that are nearly as deep as the Brazos River, the largest river west of the Mississippi that erodes, fills, and abandons its valley during each eustatic cycle (Simms et al., 2006b; Mattheus et al., 2007; Taha and Anderson, 2008). Drill cores from these compound valleys at modern estuarine locations reveal that they are filled mainly with Holocene sediments, indicating that valley-fill deposits from earlier eustatic cycles were purged prior to Holocene time. Continual excavation of valley-fill facies contributed large volumes of sediment to the coast, adding to the amount delivered by rivers and eroded from the inner continental shelf. The evidence for this is seen in the large, fluvial-dominated deltas that formed on the continental shelf (Abdulah et al., 2004; Banfield and Anderson, 2004; Wellner et al.,

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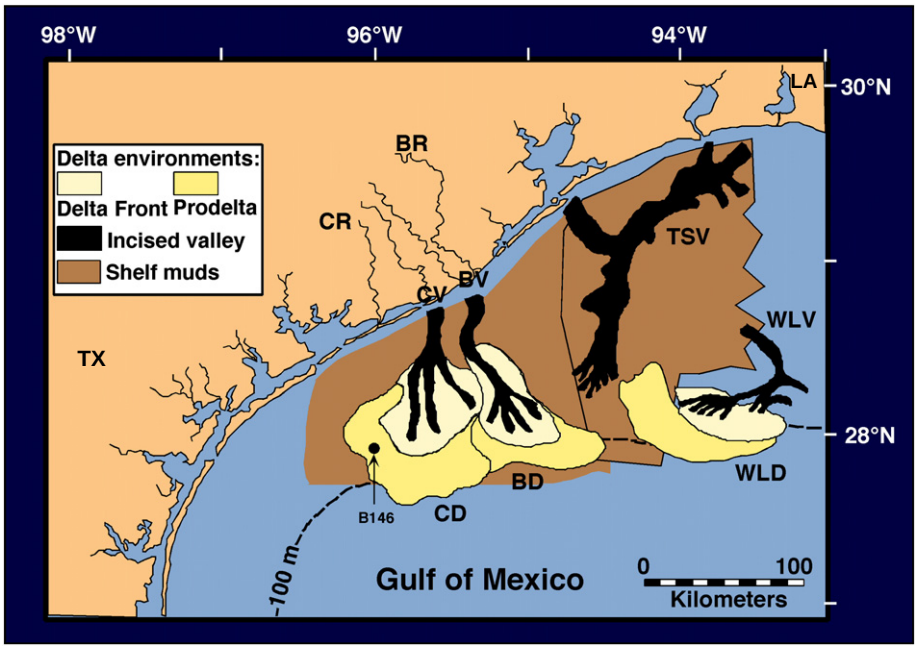

Fig. 3. Paleogeographic map showing the locations of the Colorado, Brazos, Trinity/Sabine and western Louisiana Deltas from 80,000 to 22,000 years ago. Modified from Anderson et al., 1996; Anderson, 2007. The yellow depicts sandy channel and mouth bar and prodelta deposits based on seismic facies and offshore platform borings (see Anderson et al., 2004 for greater detail). CR = Colorado River, CV = Colorado Valley, CD = Colorado Delta, BR = Brazos River, BV = Brazos Valley, BD = Brazos Delta, TSV = Trinity/ Sabine Valley, WLV = Western Louisiana Valley, WLD = Western Louisiana Delta, TX = Texas, and LA = Louisiana.

2004; Fig. 3). Seismic facies and platform borings indicate that all of these deltas had extensive sandy mouth-bar facies that were important sources of sand for the evolving coast during the subsequent transgression.

Although the rate of sea-level rise was high during the late Pleistocene transgression (10 mm/yr from ~17 ka to ~11.7 ka), preservation of coastal deposits on the outer shelf was minimal. This was likely due to deep shoreface ravinement processes resulting from high outer shelf gradients and low rates of shoreline transgression. By ~7.5 ka, the rate of sea-level rise in the Gulf of Mexico had decreased to an average rate of 1.4 mm/yr (Fig. 2), but the record of coastal evolution on the continental shelf remains fragmentary. The exceptions to this are the incised river valleys that, in the case of smaller river valleys, contain a record of flooding and associated bay evolution (Nelson and Bray, 1970; Thomas and Anderson, 1994; Simms et al., 2006b).

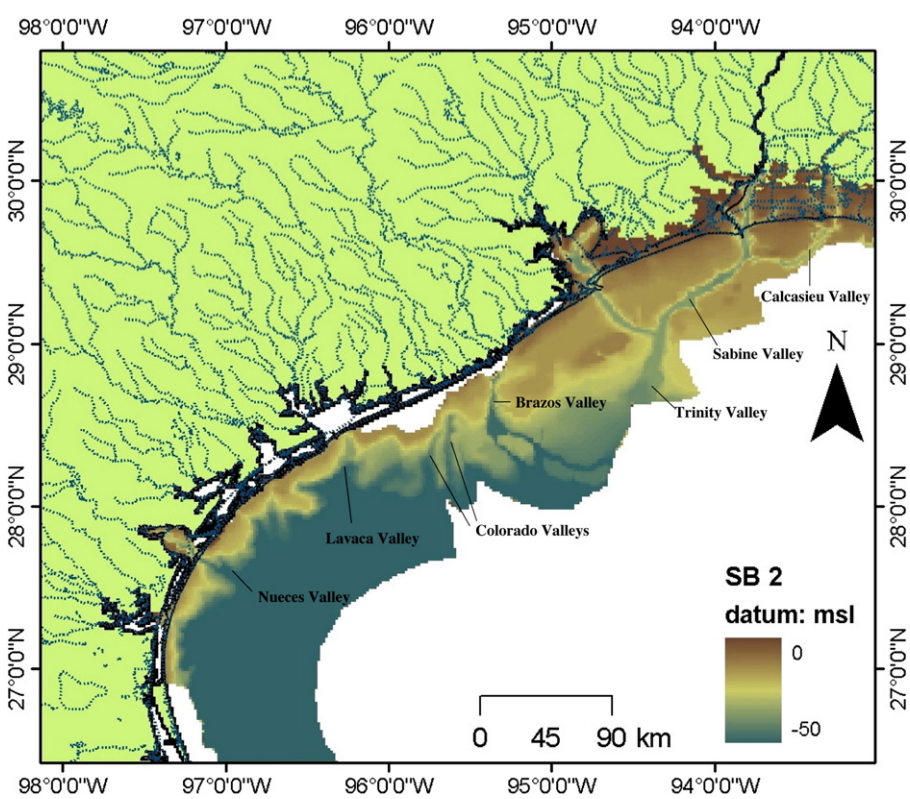

Fig. 4. Digital elevation map of the SB2 exposure surface of the northwestern Gulf of Mexico continental shelf showing fluvial valleys (modified from Simms et al., 2007a).

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All of the incised valleys of the region exhibit significant widening within a few tens of kilometers of the coast due to the steeper gradient of the continental shelf, tributary systems merging with the main trunk valley that increased discharge, and bay ravinement processes (Mattheus and Rodriguez, 2011). Landward of the modern bayline, the valleys are filled primarily with fluvial sediments. Seaward of this point they are filled with a mix of fluvial, estuarine and marine deposits (Simms et al., 2006b). It is in the lower reaches of these valleys where most sediment accommodation exists and where the interaction of sediment supply and sea-level rise prominently manifests itself in the stratigraphic record (Anderson et al., 2008; Maddox et al., 2008; Milliken et al., 2008b, 2008c; Simms et al., 2008; Taha and Anderson, 2008; Simms et al., 2010; Troiani et al., 2011). The exceptions to this are the Brazos, Colorado and Rio Grande valleys, which are filled entirely with fluvial sediment (Simms et al., 2006b; Taha and Anderson, 2008). The Holocene sediment fill of their incised valleys is composed of mostly floodplain silts and clays, with lesser amounts of sand within channels and splay deposits. Only the Brazos and Colorado valleys have been studied in detail, but aggradation of these valleys kept pace with sea-level rise (Blum and Aslan, 2006; Taha and Anderson, 2008; unpublished data from the lower Colorado valley).

6. Holocene coastal evolution

6.1. Bays

Thomas and Anderson (1994) first studied the stratigraphic architecture of incised valley fill in the offshore Trinity and Sabine River valleys (Fig. 4). They observed that these valleys exhibit an evolution that was punctuated by landward shifts in valley-fill sedimentary facies (fluvial/bayhead delta/open bay/marine) separated by prominent flooding surfaces. They interpreted this back-stepping character as being due to episodic sea-level rise, based on the observation that the same flooding surfaces offset tidal inlet and bayhead delta deposits. However, their study lacked age constraints needed to test that hypothesis and bracket the timing of backstepping (flooding) events.

Later research was partly aimed at testing the hypothesis that episodic sea-level rise caused the back-stepping stratigraphic architecture described by Thomas and Anderson (1994) and focused on obtaining chronostratigraphic data needed to establish the timing and duration of flooding events. Those studies included most of the major bays of the NGC, including Calcasieu Lake, Sabine Lake, Galveston Bay, Matagorda Bay, Corpus Christi Bay, Copano Bay, and Baffin Bay (Rodriguez et al., 2004; Anderson et al., 2008; Maddox et al., 2008; Milliken et al., 2008b, 2008c; Simms et al., 2008, 2010; Troiani et al., 2011). The bays were studied using similar methodologies in order to obtain comparable results. This included detailed mapping of the morphologies of the incised valleys using new and existing (USGS) seismic data, followed by detailed seismic facies and stratigraphic analysis aimed at mapping bay-fill sedimentary facies and surfaces bounding these facies. Seismic interpretations were corroborated using drill cores and vibracores. The bay-head deltas of these bays all experienced abrupt landward shifts that are manifest as flooding surfaces. These surfaces are conspicuous in seismic records and sediment cores (Fig. 5). A series of drill cores were collected along each valley axis to examine the flooding history and radiocarbon dating focused on constraining the ages of flooding surfaces.

Maps of valley morphology provided a framework for inferring the stratigraphic position and age of flooding events that resulted from changes in the antecedent topography of the valley (Rodriguez et al., 2005). This work demonstrated that this was indeed an important factor controlling bay evolution. In particular, inundation of relatively flat fluvial terraces along the flanks of some valleys resulted in rapid bay expansion and the formation of flooding surfaces. But not all flooding surfaces are associated with changes in valley morphology. The remaining surfaces were assumed to represent times when the rate of sea-level rise exceeded the sediment supply necessary to fill the accommodation created by sea-level rise. Unfortunately, the existing Holocene paleoclimate records for the study area are too low resolution to help constrain climate events that may have contributed to variations in sediment supply. We also found that flooding surfaces that can be shown to offset both tidal inlet/delta deposits and bayhead deltas are rare. This is partly because most of the early and mid-Holocene tidal deltas are located in the offshore part of the valley where seismic coverage and radiocarbon age constraints are limited. By late Holocene time the evolution of tidal inlet/delta deposits between bays was highly diachronous owing to differences in the time of formation of barriers and associated inlets.

Given these limitations, we took another approach to address the cause of individual flooding events. The underlying assumption used in this approach is that flooding events that are correlative between bays, and therefore regional in extent, were caused by accelerated sea-level rise, whereas more localized flooding events are attributed to reductions in sediment supply or differences in the size and shape of the valley being flooded. The limitation here comes from uncertainties in the carbon reservoir, which is known to vary across the area and can be as high as ~760 years (Aten, 1983; Kibler, 1999; Simms et al., 2007b, 2008; Milliken et al., 2008a). Even with radiocarbon reservoir uncertainties, the timing and duration of individual flooding events is constrained to within one to two centuries using current knowledge of the spatial and temporal variability in the radiocarbon reservoir and radiocarbon ages above and below flooding surfaces at two or more locations along the axis of each bay (Fig. 6).

Many of the bays we studied experienced rapid flooding from the time of their initial formation, between ~9.8 and ~9.5 ka until ~6.8 ka (Fig. 6). Several flooding events appear to be contemporaneous, and are therefore candidates for rapid sea-level change. These flooding events occurred at ~9.8–9.5 ka, ~8.9–8.5 ka, ~8.4–8.0 ka, ~7.9–7.5 ka, and ~7.4–6.8 ka (Figs. 6, 7; Anderson et al., 2010) in Calcasieu Lake, Sabine Lake, Galveston Bay, and Matagorda Bay. Corpus Christi Bay, Copano Bay and Baffin Bays also experienced backstepping events at ~9.8–9.5 ka and ~8.4–8.0 ka but many of the other events are only manifested geochemically and not stratigraphically in these more southern bays (Simkins et al., 2012).

After ~6.8 ka, flooding events were more localized (Fig. 7). This is partly due to the shapes of the valleys, which tend to widen upwards with flatter valley flanks, amplifying the magnitude of flooding events. But, this was also a time of climate variability along the western Gulf Coast, characterized by fluctuations between cool and wet to warm and dry conditions (Toomey et al., 1993; Nordt et al., 2002). This climate variability likely resulted in variations in sediment supply to bays and triggered flooding events.

6.2. Sand supply from rivers

As discussed above, flooding of most river valleys to create bays had occurred by the early Holocene, so sand supply from these fluvial sources to the evolving coast was minimal. However, the Brazos, Colorado and Rio Grande Rivers continued to supply sediment to the coast throughout the Holocene. Indeed, all three rivers formed extensive Holocene deltas (Abdulah et al., 2004; Banfield and Anderson, 2004). These deltas all lie above the MIS2 exposure surface, which is the basis for their having formed during the transgression. The water depths in which the Brazos and Colorado Deltas occur suggests that they formed between ~11.0 and 9.5 ka, but the Rio Grande Delta appears to have formed later in the Holocene. In the case of the Brazos Delta, this age is supported by a single radiocarbon age (Abdulah et al., 2004). This corresponds to a time of relatively rapid sea-level rise, so growth of these deltas implies increased sediment supply, which was likely triggered by climate change that influenced the entire area.

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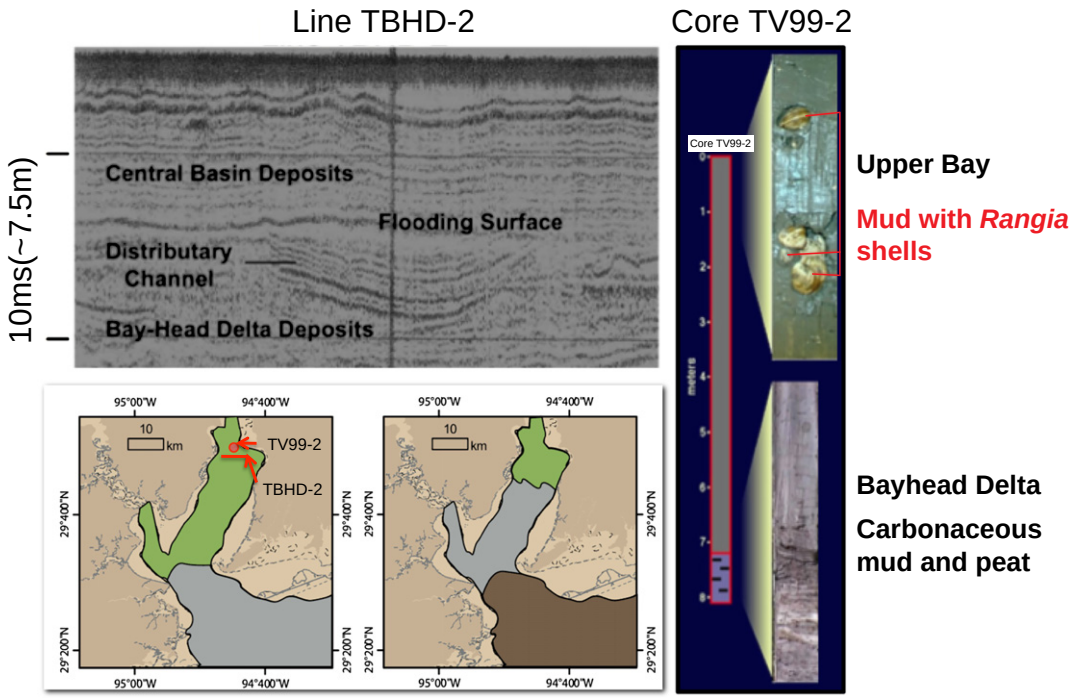

Fig. 5. Example of seismic section and core from upper Galveston Bay illustrating flooding surface and environment changes across this surface. In the seismic section the flooding surface separates subaerial delta plain deposits and the associated distributary channel below from bay deposits above this surface. Core TV99-2 sampled these facies and provided shell material for radiocarbon dating these deposits. The map depicts the extent of environmental change before and after this flooding event (modified from Anderson et al., 2008) (gray represents middle bay, green represents bayhead delta and brown represents tidal delta environments).

In central and south Texas, a major event that occurred during the late Holocene was the formation of the Texas Mud Blanket, which spread as a sheet of marine mud across the shelf (Weight et al., 2011). Evidence suggests that the mud blanket records an increase in sediment delivery to the coast by the Brazos and Colorado Rivers that resulted from the decrease in sea-level rise that allowed these rivers to fill accommodation within their valleys and begin delivering more sediment to the coast.

6.3. Sand supply from transgressive ravinement

The record of Holocene coastal evolution on the continental shelf is fragmentary compared to that of area bays. This is because subsidence was minimal during the relatively short interval of the transgression, so ravinement removed virtually all deposits above the level of storm wave influence (Rodriguez et al., 2001).

Within the study area, transgressive ravinement erodes to the toe of the shoreface, removing the upper ~8 to 10 m of the sediment column (Siringan and Anderson, 1994; Rodriguez et al., 2001; Wallace et al., 2009, 2010). The depositional environment interpretations from numerous core transects from the region show modern shoreface deposits above this water depth and marine mud resting unconformably on Pleistocene or older Holocene sediments seaward of this water depth. The efficiency of transgressive ravinement is illustrated by the fact that, with few exceptions, sand is lacking seaward of the toe of the shoreface (Siringan and Anderson, 1994; Rodriguez et al., 2001). Bathymetric highs are particularly susceptible to ravinement, unless they can be rapidly submerged below storm wave base. For example, the Brazos wave-dominated delta, which historically extended offshore into ~20 m of water, was removed in less than three decades after diversion of the river in 1929 (Rodriguez et al., 2000a).

While transgressive ravinement has removed much of the sedimentary record of Holocene coastal evolution from the continental shelf, this process was largely responsible for nourishing Holocene shorelines with sand. By far, the largest source of sand for the western Louisiana and east Texas coasts came from ravinement of the falling-stage Western Louisiana Delta (Wellner et al., 2004) (Fig. 3). Ravinement of the falling-stage Brazos and Colorado Deltas (Abdulah et al., 2004) were also important sources of sand for the central Texas coast and the Rio Grande Delta was a major sand source for the south and central Texas coast. Seismic facies mapping indicated that these were fluvial-dominated deltas with clinoform top-set facies dominated by sandy mouth bar deposits (Abdulah et al., 2004; Banfield and Anderson, 2004; Wellner et al., 2004). This is also substantiated by platform borings that sampled sand in the upper parts of these deltas.

Weight et al. (2011) calculated ravinement sediment production rates (km^3/ka) in 1000 year time increments for the area bounded by 26.5° N in the south, to 95° W, which includes the ancestral Brazos and Colorado Deltas. They performed these calculations in order to identify periods of time when large amounts of sediment were being generated by this process as the coastline moved landward. Sediment production rates were calculated using the area on the deltas, the sea-level curve of Simms et al. (2007b) and a maximum depth of transgressive ravinement of 10 m (Rodriguez et al., 2001). The calculations of Weight et al. (2011) revealed a two-order-of-magnitude increase in production from 0.4 km^3/ka to 41 km^3/ka (flux from 0.7×10^6 t/y to 69×10^6 t/y) at ~9 ka. This was followed by a gradual reduction in the volume of sediment generated by transgressive ravinement to almost zero between ~7.5 and ~4 ka as the shoreline migrated landward of these deltas. The estimated total sediment production from ravinement of these deltas was ~61.0 km^3. Assuming 60% of the eroded sediment was sand, a conservative estimate based on seismic facies and platform borings, this yields a total volume of ~36.6 km^3 of sand that was supplied to the evolving shoreline through this process.

Using the same method as Weight et al. (2011), we calculate ~18.8 km^3 of sand was eroded by ravinement of the Western Louisiana Delta. Here again, the results of seismic facies analysis and platform borings indicate that the upper portion of the delta was composed of sandy mouth bar deposits, so we use a value of 60% sand within the deltaic

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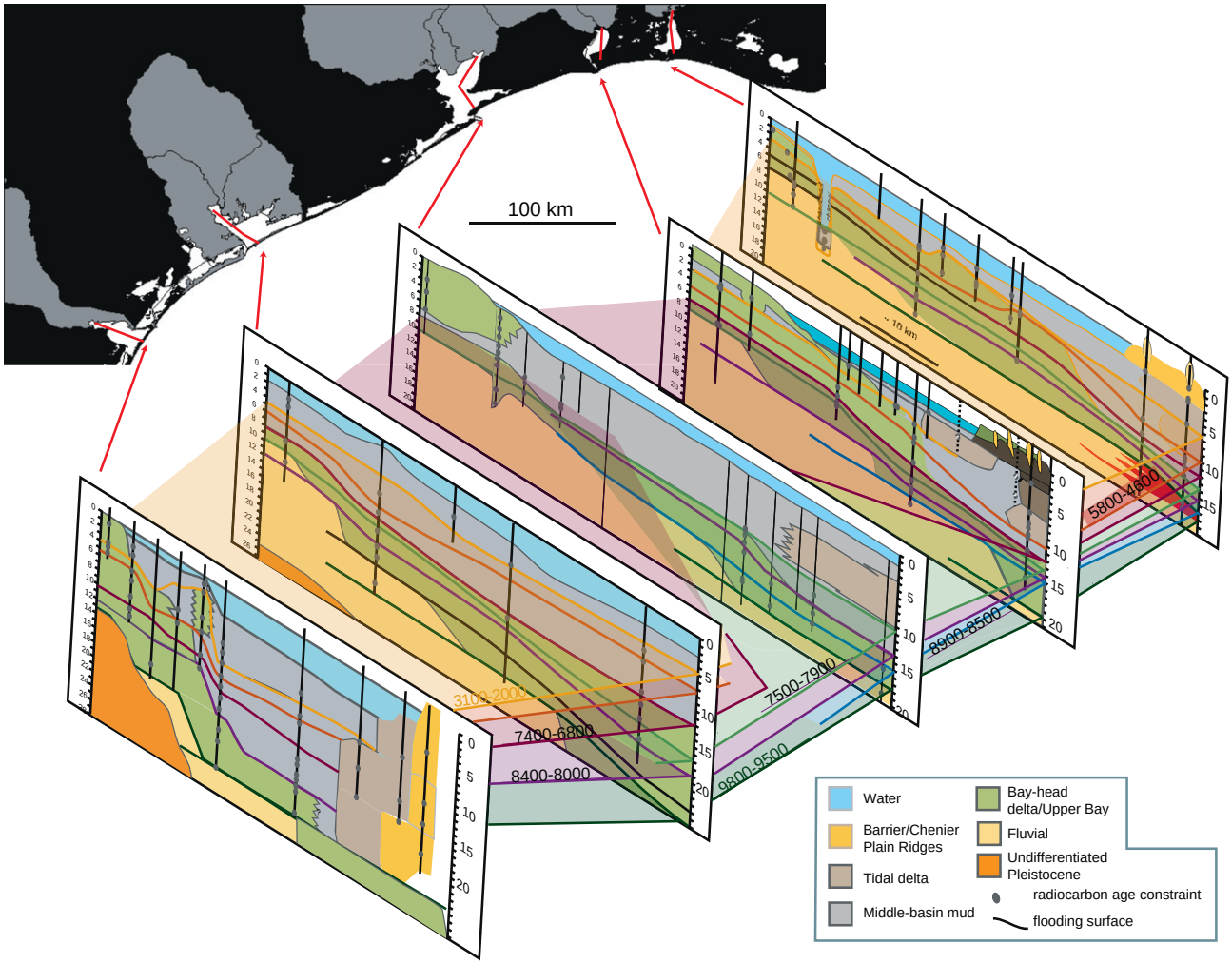

Fig. 6. Summary of Holocene bay evolution showing possible contemporaneous flooding events, designated with colored lines and numbers indicating age limits from radiocarbon ages (cal yr B.P.) of sediments above and below these surfaces (modified from Anderson et al., 2010). Depths are in meters. See Anderson and Rodriguez, 2008 for all radiocarbon dates.

deposits that were eroded. Much of this sand was deposited within the Trinity and Sabine River valleys, which both contain thick and extensive tidal inlet/delta deposits and attest to high sand supply throughout the early through mid-Holocene (Thomas and Anderson, 1994; Anderson et al., 2008).

6.4. Ancestral barriers

Several large sandy banks exist on the east Texas shelf along the flanks of the Sabine and Trinity valleys (Sabine, Heald and Shepard Banks; Fig. 8). Detailed study of these banks revealed seismic and lithofacies indicative of overstepped barrier islands (Rodriguez et al., 1999). This includes a basal unit showing landward progradation in seismic profiles and composed of interbedded sand and mud with back-barrier mollusks (Fig. 8). A sharp flooding surface separates this basal unit from a younger seaward dipping unit that is also composed of interbedded sand and mud but contains an offshore mollusk assemblage. The tops of the banks are covered by sand waves and ripples composed of sand and shell hash with a sharp flooding surface separating these deposits from the underlying units. This upper most unit is interpreted as indicative of reworking by marine processes.

Radiocarbon dates constrain the ages of the ancestral barriers associated with Sabine and Heald Banks (Rodriguez et al., 1999). The ages for back-barrier facies of Heald Bank range from ~8.4 to 7.5 ka and indicate that the barrier was rapidly overstepped thereafter. Back-barrier deposits of Sabine Bank yield ages in the range of 5.3 to 4.7 ka, and the barrier was also rapidly overstepped thereafter. These overstepping events are consistent with the subsequent formation of Galveston Island and Bolivar Peninsula, respectively. Galveston Island began to form and was prograding around the same time as a barrier island existed at the location of Sabine Bank (Bernard et al., 1959; Rodriguez et al., 2004).

6.5. Chenier Plain evolution

The Chenier Plain of western Louisiana (Fig. 1) has been studied extensively for nearly 80 years (Howe et al., 1935; Russell and Howe, 1935; Price, 1955; Byrne et al., 1959; Gould and McFarlan, 1959; Frazier, 1967; Otvos and Price, 1979; Penland and Suter, 1989; McBride et al., 2007). General consensus among early workers was that the chenier plain formed by alternating suspended sediment deposition and wave erosion of sandy mud (Howe et al., 1935; Russell and Howe, 1935) and that these changes were influenced by avulsion of the Mississippi River and its delta (Frazier, 1967). Penland and Suter (1989) demonstrated that the modern cheniers formed after ~2.5 ka and that their formation was strongly influenced by lobe shifting within the Lafourche Delta. McBride et al. (2007) focused on the variability in chenier development along the coast that reflects different factors that influence chenier development at decadal to century time scales.

Seismic records acquired and examined by Milliken et al. (2008b) imaged a prominent flooding surface at the base of the Calcasieu Chenier Plain that extends to the upper part of Calcasieu Lake. Radiocarbon ages were used to constrain the age of this flooding surface as

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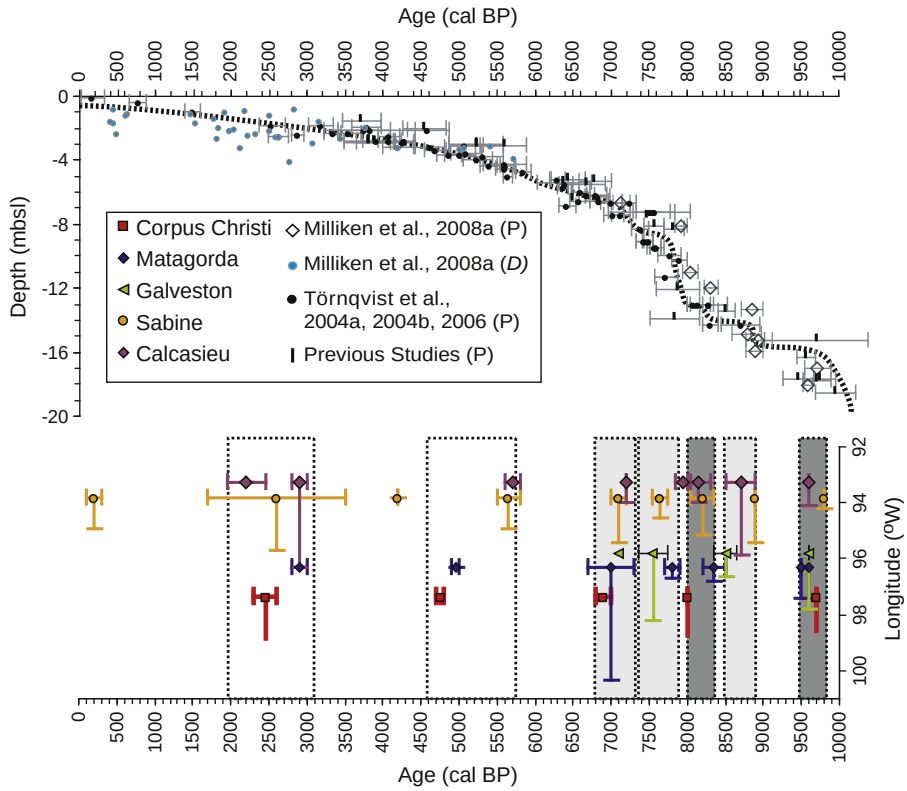

Fig. 7. Summary figure showing bay flooding surfaces and sea-level changes (data from Anderson and Rodriguez, 2008). The nonlinear regression sea-level curve (top) takes into account both horizontal and vertical errors, some of which are shown (for full ranges of error, see Milliken et al., 2008a). Episodic rates of rise are indicated in the Early Holocene. Bottom colored symbols show bay flooding events. The vertical bar reflects the magnitude of flooding events as distance of upvalley bayhead delta migration and the horizontal bar reflects uncertainty in the age constraints. Note that the frequency and magnitude of flooding events was greater during the Early Holocene, when more episodic sea-level rise occurred. (P) = Peat; (*D*) = *Donax* sp.

having formed after 2.8 to 3.0 ka. An age of ~2.6 ka was obtained below the basal part of the Chenier Plain. Thus, formation of the modern Chenier Plain followed this flooding event. In Sabine Lake, a flooding surface also occurs below the modern Chenier Plain. A single radiocarbon age of 3020 cal yr BP was acquired from estuarine sediments beneath this surface (Milliken et al., 2008c), so the two flooding surfaces are likely contemporaneous.

6.6. Evolution of modern peninsulas and barrier islands

Bolivar Peninsula formed slightly seaward of its modern location ~2.5 ka and began laterally accreting shortly after its formation (Morton, 1994; Rodriguez et al., 2004). As the peninsula accreted westward over the ancestral Trinity River incised valley, its thickness increased from 3 to 15 m and its rate of growth decreased (Fig. 9). Two distinct sets of beach ridges mark different episodes of barrier evolution. An older set curves toward the north at the west end of the peninsula, reflecting the migration and infilling of the tidal inlet (Fig. 9). A second younger beach ridge set trends parallel to the modern shoreline and curves slightly seaward toward the west. This beach ridge set is associated with the existence of a prominent ebb tidal delta that existed prior to construction of the Houston Ship Channel and associated jetties. The modern beach ridges within this younger set reflect progradation that has occurred since the construction of the jetties. Radiocarbon ages from *Donax* shells, a genus that lives in the foreshore environment, collected from the older beach ridge set range from ~1.5 to 1.2 ka (Rodriguez et al., 2004). *Donax* shells from the younger beach ridge set yielded ages younger than ~0.8 ka. This hiatus in beach ridge formation is interpreted as resulting from erosion of the seaward portion of the peninsula during a major hurricane that occurred sometime around 800 years ago followed by a phase of healing and rapid shoreline progradation (Rodriguez et al., 2004; Anderson, 2007). Three upper shoreface to inner shelf transects of cores from offshore Bolivar Peninsula all sampled marine mud onlapping the lower shoreface, indicating a recent history of transgression and retreat of the peninsula (Rodriguez et al., 2001; Fig. 10).

Galveston Island ranges in thickness from 12 m to 2 m from east to west, which reflects the relief on the underlying Pleistocene surface and the ancestral Trinity River incised valley (Fig. 4). The island has a well-documented history of progradation and westward accretion that began ~5.5 ka (Bernard et al., 1959, 1970; Morton, 1994; Rodriguez et al., 2004). Growth of the island ended after ~1.8 ka, the age of the youngest beach ridge on the island (Rodriguez et al., 2004). Since that time, the eastern end of the island has remained relatively stable and a large ebb tidal delta existed in the area (Siringan and Anderson, 1993). This is attributed to its location near the Trinity River incised valley that was the likely source of sand to the barrier. The western part of the island reached its near present day shoreline by at least ~3.0 ka (Bernard et al., 1970; Rodriguez et al., 2004). Evidence suggests that Galveston Island began eroding significantly when San Luis Pass Tidal Delta formed ~2.1 ka (Wallace and Anderson, 2013), which is also consistent with the time the east end of the island stopped prograding. This lateral change from a stable east end of the island to a retreating west end of the island corresponds to a dramatic change in the thickness of the island and hence the volume of sand that is being exhumed during transgression.

Follets Island is a thin transgressive barrier that ranges in thickness from 1 to 4 m. It rests on paleo-Brazos River and delta sediments that have yielded radiocarbon ages as young as ~3.0 ka (Bernard et al., 1970; Morton, 1994). Radiocarbon ages from the barrier itself indicate that it existed in its current location since ~2.5 ka (Wallace et al., 2010). Washover deposits dominate the backside of the island. This is one of the few locations along the coast where rates of washover are

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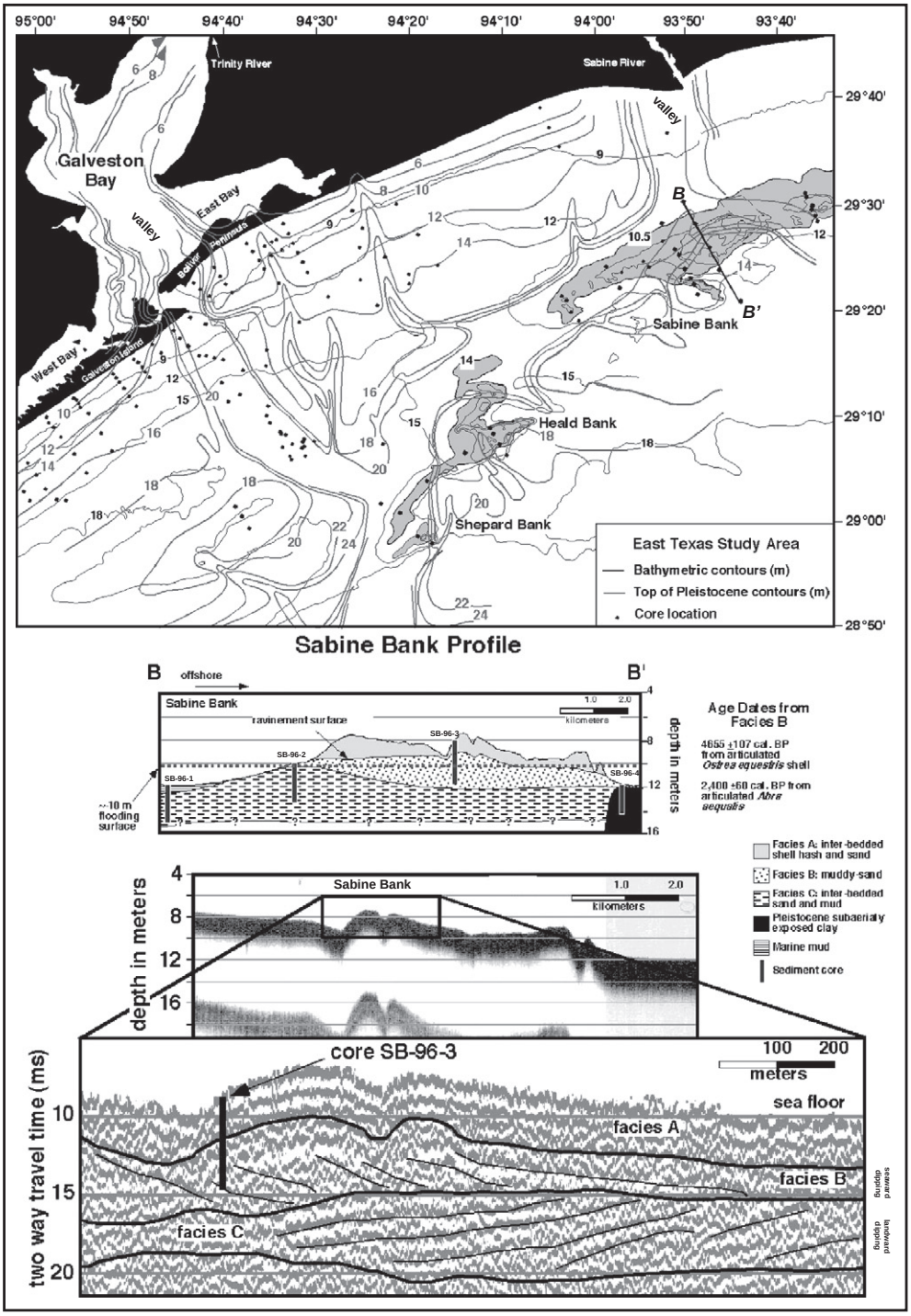

Fig. 8. Map showing locations of Trinity and Sabine incised valleys and locations of Sabine, Heald and Shepard Banks (modified from Rodriguez et al., 2004). Also shown are profiles across Sabine Bank (B) illustrating sediment facies sampled by cores, a bathymetric profile across the bank and a corresponding high-resolution (boomer) seismic profile (modified from Rodriguez et al., 1999, 2001, 2004).

consistent with a "rollover" stage of barrier island evolution and the bayline on the backside of the island is migrating landward at a similar rate to Gulf shoreline erosion.

Matagorda Peninsula is a long, narrow peninsula that extends across the mouth of Matagorda Bay (Fig. 1). Wilkinson and Basse (1978) provide radiocarbon dates which indicate that the barrier formed sometime after ~6.7 ka. This is associated with a landward shift in bay environments within Matagorda Bay at about this time (Maddox et al., 2008). Matagorda Peninsula is estimated to be at least ~3.0 ka in age and has been retreating for ~1.5 ka (Wilkinson and McGowen, 1977). Sediment cores from offshore Matagorda Peninsula indicate a history of growth throughout the Holocene (Rodriguez et al., 2001) (Fig. 10).

Matagorda Island is 10 to 14 m thick. The island formed ~4.0 ka and has been prograding since formation (Wilkinson, 1975). Sediment core transects off Matagorda Island also indicate a history of progradation during the late Holocene (Rodriguez et al., 2001) (Fig. 10).

Borings through San José Island showed that it is between 13 and 19 m thick (Shepard and Moore, 1955; Morton and Amdurer, 1974).

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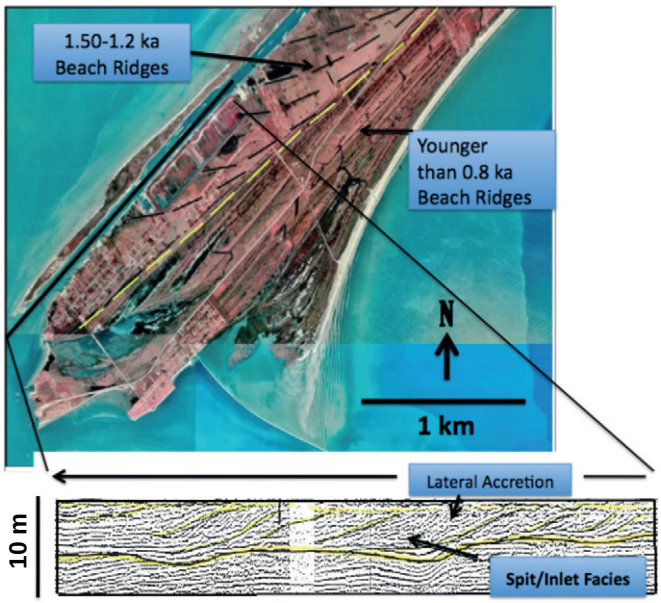

Fig. 9. Image of the west end of Bolivar Peninsula with beach ridges highlighted with black lines and two sets of distinct beach ridges demarcated with yellow dashed line. Also shown is a seismic profile collected within the Intercoastal Waterway on the backside of the western end of the peninsula showing lateral accretion and deepening of the inlet toward the west and over the Trinity incised valley (modified from Rodriguez et al., 2004; Anderson, 2007).

We recently acquired radiocarbon ages from a core transect across the island that show a progradational history extending back at least to ~2.4 ka, but did not sample the oldest part of the island. Offshore sediment core transects indicate a late Holocene history of progradation for San José Island (Rodriguez et al., 2001) (Fig. 10).

A detailed study of Mustang Island showed that it was located near its current location by ~7.5 ka, with one radiocarbon age suggesting as long ago as ~9.5 ka (Simms et al., 2006a). The 9.5 ka age was soon after the ancestral Brazos and Colorado Deltas were overstepped and about the time of the first recorded flooding event in area bays (Fig. 7), implying a period of rapid shoreline migration across the inner continental shelf. Throughout the island's history, tidal inlets shifted within nearly the same location, leaving 10–15 m thick deposits of clean, well-sorted, quartz sand. Stabilization and aggradation of the island throughout most of the Holocene was made possible by an abundant supply of sand from three sources: Pleistocene deltas and headlands, the transgressive Colorado River Delta, and the MIS 3 shoreline of the central-Texas shelf. During the late Holocene (~3.5 ka to Present), these sand sources were buried beneath the Texas Mud Blanket, which accumulated 172 km^3 (about 57% of its total volume) due primarily to a corresponding increase in shelf sediment delivery by the Brazos, Colorado, and Mississippi Rivers (Weight et al., 2011). Since this time, sand supply to the central Texas coast was restricted to sand delivered by longshore currents. Offshore core transects indicate a recent history of shoreface progradation (Rodriguez et al., 2001) (Fig. 10).

North Padre Island formed at least by ~3.7 ka but most likely around 5.0 ka (Fisk, 1959), and has prograded and transgressed throughout its history (Simms et al., 2006a; Garrison et al., 2010). The older age of ~5.0 (possibly as old as 5.5 ka) is supported by a change to hypersaline conditions within Baffin Bay around this time suggesting isolation of the bay from the Gulf of Mexico (Simms et al., 2010). A period of progradation of the island is supported by offshore core transects that show shoreface deposits prograding seaward over shelf mud (Rodriguez et al., 2001) (Fig. 10).

South Padre Island is at least ~2.0 ka old, but is likely much older, based on back-barrier sediment cores with washover deposits from the island that date back to over ~5.0 ka (Wallace and Anderson, 2010).

6.7. Historical changes

Historical shoreline changes have been well documented in western Louisiana (Penland et al., 1992) and in Texas (Morton, 1979; Paine et al., 2012) and this monitoring continues today. Shoreline change in western Louisiana has been highly variable, ranging from +4.1 m/yr near the Sabine Chenier Plain to −1.7 m/yr to the east of the mouth of Lake Calcasieu, with considerable spatial variability over decadal time intervals (Penland et al., 1992; Draut et al., 2005). The eastern portion of the Chenier Plain in Vermillion Parish is experiencing mudflat progradation. In general, the central Chenier Plain in Cameron Parish is in an erosional, chenier-forming stage and the western part in western Cameron Parish is experiencing slow growth and beach-ridge formation (Owen, 2008). McBride et al. (2007) present a six-stage geomorphic process-response model that explains this variable chenier-plain evolution primarily as a function of: (1) the balance between sediment supply and energy dissipation; (2) local sediment reworking and lateral transport; and (3) tidal-entrance dynamics.

In Texas, Gulf shoreline change rates also vary along the coast and have averaged approximately −1.6 m/yr in upper Texas, –0.30 m/yr in central Texas, and −1.9 m/yr in south Texas (Gibeaut et al., 2006; Paine et al., 2012). Net shoreline retreat has occurred along 84% of the Texas Gulf shoreline, resulting in an estimated land loss of 13,890 acres since 1930 (Paine et al., 2012). Bay shoreline changes are even more extreme and spatially variable. Severe storms also exert considerable influence on shoreline erosion and a single storm can easily result in an order of magnitude increase in erosion rates.

Tidal inlets in Texas have also experienced significant changes during historical time, including inlet migration of up to a kilometer or more, changes in channel depth, and associated changes in the size of both ebb and flood tidal deltas.

6.8. Anthropogenic influence

There are many examples of human-induced coastal change in the study area (Wallace et al., 2009). It is beyond the scope of this paper to summarize all of these, so we will limit our discussion to those events that had wide-reaching impacts and that help us compare natural versus anthropogenic influences on coastal evolution, both in terms of rates and magnitudes of changes in shoreline position.

The natural flow and sediment transport of rivers in the area has been significantly altered by the construction of dams and excessive water usage. The Brazos, Colorado and Rio Grande Rivers all formed extensive fluvial dominated deltas during the falling stage of sea level and at times during the transgression, but today only the Brazos River forms a wave-dominated delta. Excessive water demand and dams have reduced the flow and sediment delivery of the Colorado and Rio Grande Rivers to the point where they are no longer considered important sediment contributors to the coast. Indeed, the Rio Grande virtually ends at the lower end of the El Paso valley, except in seasons of above normal flow. Until 1929, the Colorado River was virtually choked by a logjam in its lower reaches that blocked much of the sediment being transported by the river (USACE, 1992). Destruction of the logjam in 1929 resulted in rapid growth of a delta within eastern Matagorda Bay, but the Colorado River no longer flows into the Gulf of Mexico because it was diverted to Matagorda Bay in 1992 (Barcak et al., 2007).

In 1929, the original course of the Brazos River was diverted by the U.S. Army Corps of Engineers to a new location ~10 km west of its 1929 location, where a relatively large wave-dominated delta existed (Fig. 11). Within a few decades that delta was eroded and a new delta formed at the new river mouth (Bernard et al., 1970; Rodriguez et al., 2000a). During the first few decades after diversion, supply from

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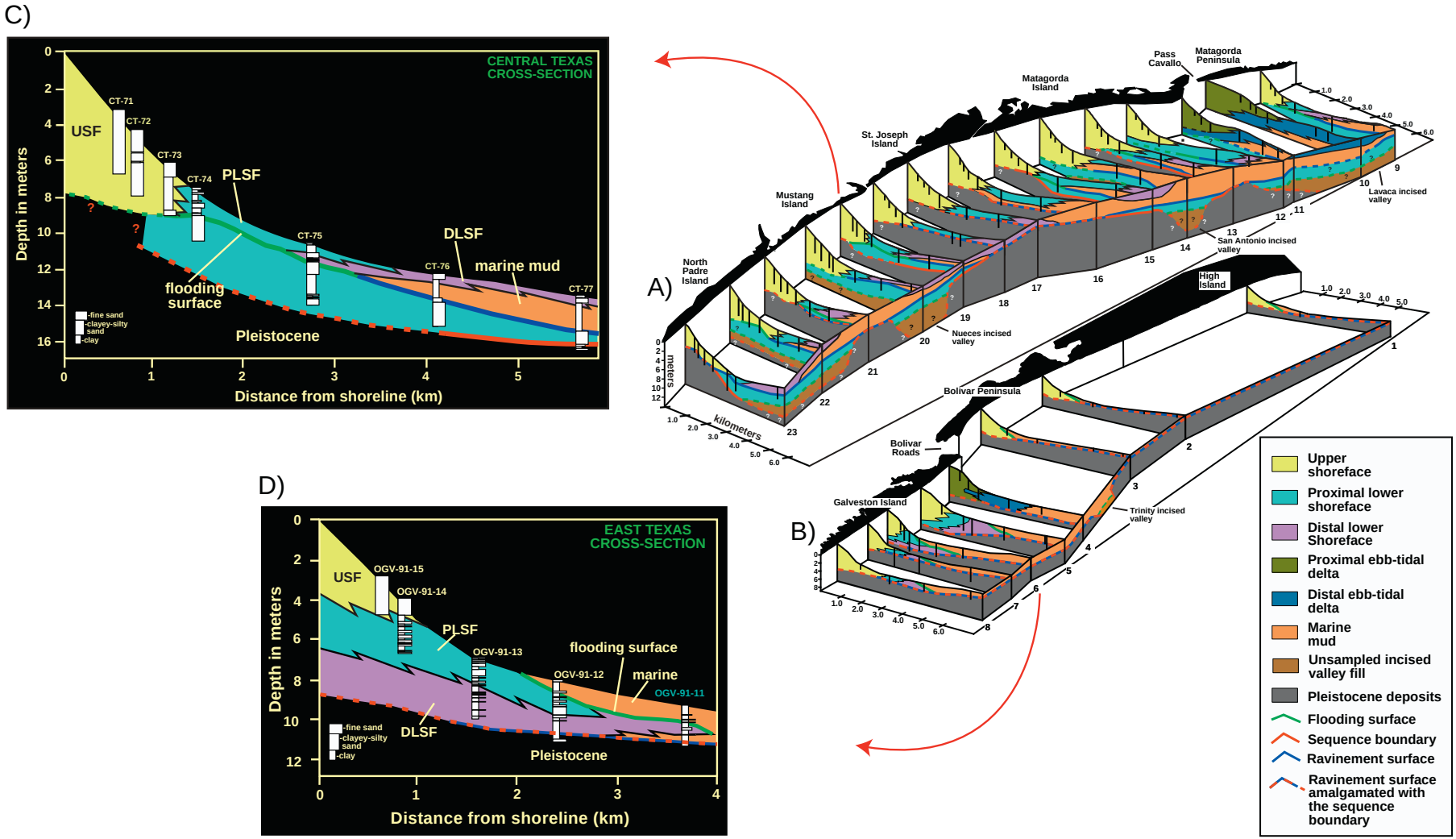

Fig. 10. Shoreface to inner shelf core transects from central (A) and eastern (B) Texas (modified from Rodriguez et al., 2001). Transects from offshore of Bolivar Peninsula all show marine mud onlapping the lower shoreface, indicating landward retreat of the shoreline. Core transects of central Texas indicate more variable conditions with marine mud onlapping shoreface deposits overlain by upper shoreface deposits overlying proximal lower shoreface (PLSF) and marine deposits (C) (modified from Rodriguez et al., 2001). Note that other transects from central Texas indicate uninterrupted shoreface progradation with shoreface deposits generally resting above offshore deposits. Core transects from offshore Galveston Island also show a variable stratigraphic pattern (D) (modified from Rodriguez et al., 2001), indicating progradation at the east end of the island shifting toward a more transgressive condition toward the west.

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Fig. 11. Aerial photographs from 1930 and 1989 showing historical changes in the Brazos Delta caused by the 1929 diversion of the Brazos River (modified from Rodriguez et al., 2000a).

erosion of the former delta led to rapid growth of the new delta (Fig. 11). Once the sand supply from erosion of the pre-1929 delta was depleted, growth of the new delta decreased considerably. During the past several decades the delta has entered a new phase of episodic growth associated with pulses of sediment supply from the river. These pulses occur during and following major flooding events that are associated with decadal climate (ENSO) variability (Rodriguez et al., 2000a). Fraticelli (2006) demonstrated that each of five episodes of rapid delta growth occurred when a period of extensive drought, La Niña events, preceded an El Niño-induced flood. The droughts were instrumental in removing vegetation and pre-conditioning the drainage basin for erosion. It is not clear how this climate-controlled variability in sediment delivery to the delta was influenced by land-use changes in the drainage basin and alterations in river flow associated with the construction of numerous dams on the Brazos River and its tributaries. However, the pre-1929 delta was characterized by similar ridge and swale features that exist on the modern delta and these features are known to reflect episodic pulses of sediment supply and shoreline growth (Fig. 11).

Alteration of discharge and sediment yields of rivers that flow into area bays has also had an impact on these environments, most notably on bayhead deltas. During the late Holocene, prominent bayhead deltas formed in the upper reaches of these bays (Anderson et al., 2008; Maddox et al., 2008; Milliken et al., 2008b, 2008c; Simms et al., 2008). Throughout the late Holocene and into the late 19th Century, the Trinity River bayhead delta was stable and at times experienced growth (Anderson et al., 2008). Today, growth of the delta has slowed compared to the long-term rate of growth, but this change is small compared to rates of bay shoreline change and loss of wetlands (White et al., 2002) despite dam construction and alteration of the natural channel system of the delta (Phillips et al., 2004). The development of other bayhead deltas has also been severely hampered in historical time, and the Sabine bayhead delta has virtually disappeared (Milliken et al., 2008c).

One of the more significant impacts of humans on the coast has been the construction of jetties at the entrances to navigation channels. These features have altered the natural longshore sand transport system, effectively trapping sand adjacent to the jetties and blocking sand transport in and out of bays to alter ebb and flood tidal deltas. One of the best examples is the Bolivar tidal delta complex between Bolivar Peninsula and Galveston Island. The flood tidal delta formed in the late Holocene as Bolivar Peninsula accreted to the west to form the inlet between the two barriers. Initially, the flood tidal delta extended several kilometers into lower Galveston Bay (Rodriguez et al., 1998; Anderson et al., 2008), but with time the delta decreased in size and was buried beneath bay mud (Siringan and Anderson, 1993). Historical charts from 1867 show that a large ebb tidal delta existed offshore. After construction of the jetties the ebb tidal delta shrank in size and was largely buried beneath marine mud. Meanwhile, sand that was eroded from the ebb tidal delta contributed to shoreline progradation at the east end of Galveston Island.

7. Discussion

During the early Holocene, from ~10 to ~8.0 ka, the linear average rate of sea-level rise in the NGC was 4.2 mm/yr. We are rapidly approaching similar rates, hence this time interval provides a window of opportunity for investigating coastal response to this rate of rise.

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The record of early Holocene coastal evolution on the western Louisiana and east Texas continental shelf is incomplete. In central Texas, the Brazos and Colorado rivers both nourished large deltas (Van Heijst et al., 2001; Abdulah et al., 2004) and in south Texas there was a large delta associated with the ancestral Rio Grande River (Banfield and Anderson, 2004). The exact age of these deltas is not known, but it is likely that they formed as a result of high sediment supply associated with an extended cool, wet period in central and south Texas during the early Holocene (Toomey et al., 1993; Nordt et al., 2002). The termination of delta development occurred at ~9.0 ka when all three deltas were buried during an initial phase of growth and expansion of the Texas Mud Blanket, which is still accumulating mud (Weight et al., 2011). Thus, while the rate of sea-level rise was relatively rapid, sediment supply from the larger rivers of the area was at times able to keep pace and even produce deltas. Elsewhere and outside the direct influence of rivers, there were episodes of shoreline stability followed by rapid retreat.

The strongest manifestation of episodic coastal change occurs within area bays in the form of landward stepping bayhead deltas bounded by sharp flooding surfaces (Figs. 5, 6). This begs the question, do these surfaces reflect a threshold response to periods when sediment supply could not keep pace with the overall rate of sea-level rise or were there episodes of rapid coastal inundation caused by pulses of more rapid sea-level rise?

It is generally thought that northern Hemisphere deglaciation was nearly complete by the early Holocene, the exception being the Laurentide Ice Sheet (LIS). Thus, the LIS and Antarctic ice sheets were the principle source of the remaining ~30 m of sea-level rise (Carlson, 2011). Anderson et al. (2013b) recently summarized results of research aimed at evaluating the timing and extent of post-LGM (Last Glacial Maximum) ice sheet retreat in Antarctica and implications for sea-level rise. This review led to the conclusion that retreat of the East Antarctic, West Antarctic and Antarctic Peninsula ice sheets from the continental shelf was still occurring during the Holocene and that the retreat of these ice sheets was diachronous and mostly step-wise around the continent. They further argue that this behavior is expected given regional differences in the factors that control ice sheet behavior. These include: (1) the highly variable size of individual drainage basins and volumes of ice flowing toward the coast; (2) differences in the physiography and geology of the continental shelves on which the ice sheets advanced and retreated; (3) differences in isostatic depression and rebound; (4) potential for intrusion of warm deep water onto the continental shelf and associated erosion of the ice shelves; (5) variable climatic conditions around the continent; and (6) potential for focused subglacial water outflow leading to underpinning of ice streams. As the ice sheets retreated, their drainage became more subdivided into smaller ice streams and outlet glaciers, leading to progressively smaller and more episodic contributions to sea-level rise. This implies that Holocene sea-level rise was episodic and punctuated by rapid pulses triggered by individual ice stream collapse, as opposed to uniform ice-sheet melting.

Modern bayhead delta plains are mainly situated less than a meter above sea level, so a sea-level rise of a few decimeters will result in significant landward migration. However, even the most recent and precise sea-level curves for the Gulf Coast region lack the level of resolution needed to document rises in sea level of a few decimeters (Törnqvist et al., 2004a, 2004b, 2006; Milliken et al., 2008a; Livsey and Simms, 2013). So, we rely on the timing of flooding events as evidence that they were eustatically controlled. Our data set includes several bays that span several climate zones and whose valleys are characterized by highly variable antecedent topography, which should result in diachronous flooding histories. Where bay flooding surfaces appear to occur contemporaneously across the region, they are interpreted as having a eustatic control, although it is possible that some of these surfaces result from regionally significant and synchronous climate changes.

Our results reveal five major flooding events at ~9.8–9.5 ka, ~8.9–8.5 ka, ~8.4–8.0 ka, ~7.9–7.5 ka, and ~7.4–6.8 ka that appear contemporaneous, at least between Calcasieu Lake, Sabine Lake, Galveston Bay, and Matagorda Bay (Anderson et al., 2010), given the constraints of our radiocarbon dating (Figs. 6, 7). The magnitudes of change during each flooding event vary between bays, but this is expected given differences in the antecedent topography of the valleys and differences in sediment supply to these valleys.

The ~8.4–8.0 ka flooding event has been documented in other parts of the world and is believed to have been caused by mass wasting of the Laurentide Ice Sheet (Törnqvist et al., 2004b; Cronin et al., 2007; Kendall et al., 2008; Rodriguez et al., 2010).

In Galveston Bay, the most dramatic flooding event was the ~7.9–7.5 ka event, which resulted in a landward shift of bay-head deltas of nearly 30 km and increases in open-water area of up to 30% within two to three centuries (Anderson et al., 2008). This equates to maximum rates of bay-head delta migration during this event on the order of 150 m/yr, a rate that is unprecedented in modern times.

After ~7.5 ka, the linear average rate of sea-level rise in the northern Gulf of Mexico decreased to ~1.4 mm/yr (Fig. 2). Only one apparently regional flooding event, the ~7.4–6.8 ka event, occurred after this time (Fig. 7). Since this time, bays have undergone more localized changes that were likely the result of climate-induced reductions in sediment supply and flooding related to variations in valley morphology (Fig. 7). The coastlines of western Louisiana and east Texas continued to experience episodes of rapid landward movement, at times as fast as 20 m/yr, followed by periods of shoreline stability (Fig. 12). During this same time interval, the central Texas coast was relatively stable (Simms et al., 2006a). This was made possible by higher sand supply from converging longshore currents (Curray, 1960) and by steeper offshore and onshore gradients (Rodriguez et al., 2001). Coastal evolution in south Texas during the mid-Holocene is not well known, but the initial expansion of the Texas Mud Blanket onto the transgressive Rio Grande Delta marked a major change in coastal evolution (Weight et al., 2011).

During the late Holocene (between ~4.0 ka and historical time), the rate of sea-level rise in the northern Gulf of Mexico decreased to an average linear rate of ~0.4 to 0.6 mm/yr, well below the current rate (Fig. 2). Yet, the geological record is one of continued coastal change. It was during this time interval that the modern chenier plains of western Louisiana and most of the barrier islands and peninsulas of east Texas evolved (Fig. 13). But, chenier plain evolution would not take place until after the Calcasieu and Sabine incised valleys had been filled to within a few meters of sea level, their lower reaches having been filled mainly by large tidal deltas (Milliken et al., 2008b, 2008c). Likewise, the formation of chenier ridges led to a significant decrease in the size of these tidal deltas. Both Sabine Lake and Calcasieu Lake experienced major flooding episodes between ~2.5 and 1.8 ka, when their bayhead deltas shifted northward approximately 15 km. This was followed by progradation of these deltas to their current locations. In contrast, the Trinity Delta in Galveston Bay was located in its current location throughout this same time interval. Likewise, the Lavaca Delta in Matagorda Bay (Maddox et al., 2008) appears to have remained relatively stable during the late Holocene. Thus, the flooding event that impacted the ancestral Calcasieu and Sabine estuaries was localized in extent and likely the result of local climate change and an associated reduction in sediment supply, although the exact cause warrants further research.

The late Holocene was a time of growth of Galveston Island and Bolivar Peninsula. Associated with their growth, the Bolivar tidal inlet grew narrower and deeper and the Bolivar flood tidal delta experienced a significant reduction in size (Siringan and Anderson, 1993; Anderson, 2007). In contrast, seismic records and cores from southern Matagorda Bay show no evidence of a large flood tidal delta having existed during the evolution of Matagorda Peninsula and Matagorda Island (Maddox

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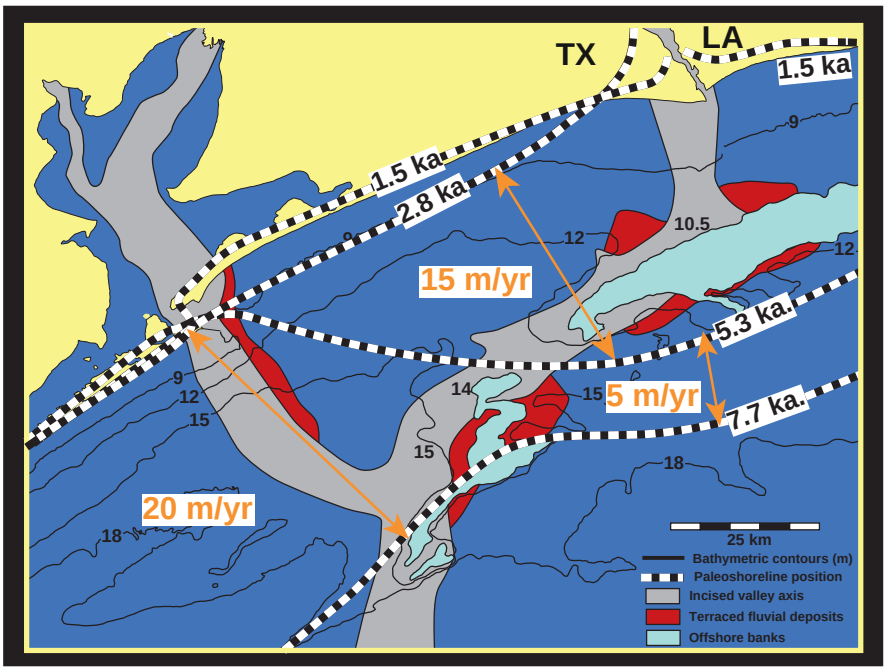

Fig. 12. Shoreline change in western Louisiana and east Texas during the middle to late Holocene (modified from Gould and McFarlan, 1959; Thomas and Anderson, 1994; Rodriguez et al., 2004). Orange arrows and numbers indicate average rates of shoreline migration.

et al., 2008). The variability reflects differences in the evolutionary histories of the barriers and inlets.

At some point in the evolution of the east Texas peninsulas and barriers they stopped growing and began retreating. The exact time of this shift from growth to erosion is constrained by varying resolutions and occurred at different times along the coast. The most common explanation for this reversal in coastal evolution is a decrease in sediment supply via transgressive ravinement once offshore sand sources were overstepped by the landward shifting shoreline (Anderson, 2007; Wallace and Anderson, 2013). The variable timing of coastal retreat reflects differences in barrier thickness, differences in localized sand supply, offshore profiles and differences in subsidence along the coast.

Current rates of Gulf shoreline erosion in western Louisiana and east Texas are approaching rates of the early and mid-Holocene (Penland et al., 1992; Rodriguez et al., 2004). Shoreline erosion in bays is also high (Wilkinson and McGowen, 1977; Morton, 1979, 1994). This is attributed mainly to diminished sediment supply and high rates of subsidence that are, in turn, due to both natural and anthropogenic causes.

Sediment supply for the Texas coast has been regulated by the availability of reworkable inner-shelf deposits, fluvial delivery, and up-drift shoreline reworking (Morton, 1994; Anderson et al., 2004). Today, the only potentially significant fluvial sediment suppliers to the coast are the Brazos and Rio Grande Rivers (Morton, 1979), although both are now minimal contributors because of damming and irrigation withdrawals (Morton and Pieper, 1975; Mathewson and Minter, 1976). Further, the advancing transgressive ravinement surface, which is located between −8 and −10 m water depths, has exhausted all large sand sources so that sand supply via this process has diminished drastically.

Regional shifts in Holocene hurricane activity has been linked to climate variability (such as El Niño/Southern Oscillation and the North Atlantic Oscillation), although this could be a reflection of regional-scale controls on hurricane intensity (Wallace et al., in press). Hurricane impacts can also influence coastal evolution, and evidence exists that impacts of storms combined with accelerated sea-level rise during historic time are leading to an acceleration of coastal change in east Texas (Wallace and Anderson, 2013). There is no evidence for increased storm activity in the Gulf of Mexico during historic time relative to the late Holocene (Wallace and Anderson, 2010), so the role of storms is mainly an exacerbation of other factors. In addition, storms account for some of the variability in long-term coastal erosion.

Changes in Holocene coastal evolution in the NGC have important implications for coastal change in the 21st Century. In particular, the overall decrease in sand supply, coupled with accelerated sea-level rise, does not bode well for the coastline. The geological record indicates that the ongoing increase in the rate of sea-level rise will severely impact low gradient coasts and indeed has already had its impact along the upper Texas Coast (Wallace and Anderson, 2013). If the overall rise is punctuated by rapid, decimeter-scale rises from ice sheet contributions, the impact will be catastrophic.

In the Gulf Coast region, bays are likely to experience the most dramatic change as their bayhead deltas are already at the tipping point where they are no longer able to keep pace with sea-level rise and either step landward or simply vanish. Furthermore, area bays have a history of dramatic response to climatically-induced changes in sediment supply and the exact nature of the climate events that led to these changes require further investigation. There is also a need for research aimed at gaining a better understanding of oceanographic and meteorological influences on decadal sea-level oscillations, which have been recorded at many tide-gauge records within the NGC (Kolker et al., 2011).

If we have learned anything from the lessons of human alteration of the coast, it is that coastal systems respond rapidly to change. The study area includes numerous examples, including (1) erosion of the old Brazos River Delta, transport of that sand by longshore currents ~10 km to the west and formation of a new delta within a few decades; (2) destruction of Bolivar tidal delta and seaward accretion of the east end of Galveston Island associated with jetty construction; (3) virtual elimination of sand delivery to the coast by the Rio Grande River; and (4) numerous beach nourishment projects that have had life histories of a few years at best.

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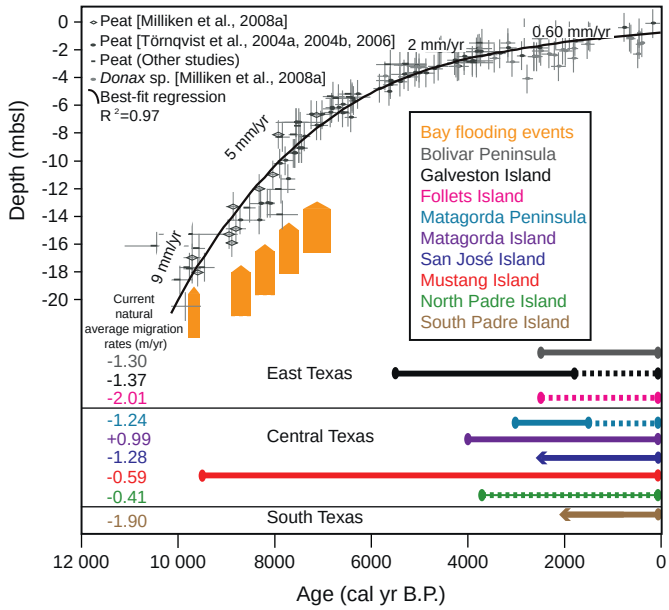

Fig. 13. Gulf of Mexico Holocene non-linear sea-level curve with Texas barrier island histories and bay flooding events (adapted from Milliken et al., 2008a [see for all included studies]; additional data from Törnqvist et al., 2004a, 2004b, 2006). These systems have had variable responses to sea-level rise over the Holocene (Anderson et al., 2010). The histories of Bolivar Peninsula (Rodriguez et al., 2004), Galveston Island (Bernard et al., 1959; Rodriguez et al., 2004), Follets Island (Bernard et al., 1970; Morton, 1994; Wallace et al., 2010), Matagorda Peninsula (Wilkinson and McGowen, 1977), Matagorda Island (Wilkinson, 1975), San José Island (this study), Mustang Island (Simms et al., 2006a), North Padre Island (Fisk, 1959), and South Padre Island (Wallace and Anderson, 2010) are shown. Stability, progradation, and/or aggradation is denoted by solid horizontal colored lines. Dashed horizontal colored lines denote transgression. Arrows for San José and South Padre Islands indicate minimum formation ages (This study and Wallace and Anderson, 2010). Bold orange arrows designate timing of bay flooding surfaces that appear to be contemporaneous across several bays (data from Anderson and Rodriguez, 2008). Also shown are the current natural average migration rates (from Texas Bureau of Economic Geology).

8. Conclusions

The Holocene evolution of the NGC was controlled by processes that were set in motion much earlier, in particular the formation of large sand bodies that would later serve as the principle sand supply for barriers and the creation of the highly irregular antecedent topography on which Holocene coastal systems evolved.

During the middle and late Holocene, the east Texas and western Louisiana coasts experienced episodes of stability and growth followed by rapid shoreline retreat, while the central Texas coast remained relatively stable (Rodriguez et al., 2004; Simms et al., 2006a). This highly variable response of coasts to sea-level rise highlights the important role of other factors, in particular sediment supply and the antecedent topography on which coastal environments formed, in coastal evolution.

Most modern coastal barriers of the upper Texas coast formed after ~5.5 ka, after the rate of sea-level rise decreased from an average of 1.4 mm/yr to between ~0.6 and 0.4 mm/yr. Regional differences in barrier evolution were regulated by differences in the rate of sediment supply and antecedent topography.

Throughout the Holocene, direct sand supply to the evolving coast by rivers was highly variable, with episodes of delta development in the early Holocene that were associated with a period of cooler and wetter climate conditions. Most of the sand that nourished the coast during the middle to late Holocene was derived from offshore sources, through transgressive ravinement of paleodeltas.

The early Holocene evolution of area bays was punctuated by rapid and apparently synchronous flooding events, which implies a eustatic control. The episodic nature of sea-level rise is attributed to spatially diachronous and episodic retreat of individual marine ice sheet drainage systems that responded independently to a number of different forcing mechanisms (Anderson et al., 2013b). Indeed, two flooding events, one between ~8.4 and 8.0 ka and the other between ~7.4 and 6.8 ka, were associated with known episodes of ice-sheet retreat in North America and Antarctica.

During the last few millennia, several barrier islands and peninsulas of the upper and south Texas coast stopped growing and began retreating. This reversal in coastal evolution was caused by a decrease in sediment supply via transgressive ravinement once offshore sand sources were bypassed by the landward shifting shoreline. Differences in the exact timing of retreat reflect differences in sand supply, barrier thickness, and the offshore profiles of these barriers. The central Texas coast has had a long history of relative stability owing to persistent sand supply by converging longshore currents and greater onshore and offshore gradients.

The current rate of sea-level rise in the Gulf of Mexico is approaching that of the early Holocene (4.2 mm/yr), but varies along the coast due to differences in subsidence. This is reflected by coastal erosion rates of the 20th to 21st centuries, which have been highly variable, even on a local scale. Overall, there is a lack of sand supply needed to sustain the Gulf shoreline and erosion is pervasive, even along the central Texas coast. Local variability in erosion rates highlights the importance of sediment supply, subsidence, storm impacts, and anthropogenic influences on coastal evolution. More localized oceanographic controls on sea-level variability may also influence coastal erosion over this time frame, but this effect is currently understudied.

Tropical storms and hurricanes, while exerting the most noticeable coastal change, mainly punctuate overall shoreline retreat. Severe storms also remove sand from the shoreline, transporting it far offshore where it is lost from the coastal transport system. Along the upper Texas coast, severe storms have resulted in minimal sand lost offshore during the Holocene (Wallace et al., 2009; Wallace and Anderson, 2013).

During historical time, humans have become the dominant control on coastal change, primarily by altering river discharge and sediment supply to the coast, increasing rates of subsidence through fluid withdrawal, and through Global Climate Change and associated accelerated sea-level rise (Anderson et al., 2013a). The current rate of erosion of some barriers is unprecedented relative to the geologic past (Wallace and Anderson, 2013), and current rates of Gulf shoreline erosion are approaching rates of the early and mid-Holocene (Rodriguez et al., 2004). Indeed, compared to natural changes that took place during the Holocene, anthropogenically induced changes are virtually instantaneous.

Acceleration of sea-level rise is not a prediction, it is reality (Engelhart et al., 2011; Kemp et al., 2011); the current rate in the northern Gulf of Mexico is about five times the pre-historic rate of the past ~4.0 ka (Törnqvist et al., 2004a, 2004b, 2006; Milliken et al., 2008a; Livsey and Simms, 2013). Likewise, coastal change due to this acceleration and other factors, both natural and anthropogenic, has already begun and in some cases is occurring at alarming rates. The geological record indicates that the ongoing increase in the rate of sea-level rise will continue to severely impact low gradient coasts at rates that, thus far, exceed the reaction time of policy makers to respond.

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